

# Product Requirements and Specifications Document

## Product Name: SEO Analyzer Chrome Extension by Thrivemattic

### 1. Introduction

#### 1.1 Overview

The SEO Analyzer Chrome Extension is designed to provide in-depth SEO analysis for any web page. This tool helps users identify SEO issues and offers recommendations to improve their website's performance and visibility on search engines.

#### 1.2 Goals

- Provide comprehensive SEO analysis
- Offer actionable SEO recommendations
- Improve page rankings on search engines
- Increase user engagement and download rate

#### 1.3 Version Information

This document specifies version 3.2 of the SEO Analyzer Chrome Extension. It represents a major revision, incorporating advanced features and a refined scoring system.

#### 1.4 User Personas

##### **Persona 1: Small Business Entrepreneur**

- Profile: Runs a successful business with at least \$20,000 in monthly recurring revenue
- Needs: Easy-to-understand SEO metrics, clear explanations of SEO concepts, prioritized action items
- Goals: Invest in digital marketing, improve website SEO

##### **Persona 2: Digital Marketing Analyst/Manager at a Startup**

- Profile: In charge of digital marketing operations and lead generation
- Needs: Detailed SEO analysis, comprehensive reports, technical insights
- Goals: Optimize website, quickly analyze web pages, provide recommendations to management/tech team

##### **Persona 3: Head of Digital Marketing**

- Profile: Responsible for overall marketing strategy
- Needs: Quick overview of SEO health, high-level insights
- Goals: Rapidly assess website optimization, make informed decisions

## 2. Functional Requirements

### 2.1 Feature List and Prioritization

#### **P0 (Must-have features):**

1. Comprehensive SEO Analysis
2. Keyword Analysis
3. Performance Analysis (Core Web Vitals)
4. Content Quality Assessment
5. Technical SEO Evaluation
6. On-Page Element Analysis
7. Scoring System with Weighted Severity Levels
8. Actionable Recommendations

#### **P1 (Important, but not critical):**

9. Media Optimization Check
10. User Experience and Accessibility Evaluation
11. Link Analysis
12. Structured Data Validation
13. Security Assessment
14. Industry Benchmarks for Comparison
15. CSV Export Feature

#### **P2 (Nice-to-have features):**

16. Compliance Scoring (GDPR, CCPA, ADA)
17. Social Media Assessment
18. Historical Tracking of SEO Scores
19. Competitive Analysis Feature
20. Custom Scoring Options for Users
21. Progress Bar for Analysis Process [New]
22. Collapsible Category Scores [New]

### 2.2 Feature Descriptions

1. Comprehensive SEO Analysis
  - Analyze various on-page SEO elements
  - Provide a detailed scoring system with weighted severity levels
  - Offer industry benchmarks for comparison

Example for Comprehensive SEO Analysis:

Description: Perform a thorough analysis of various SEO factors on a given web page.

User Stories:

- As a small business owner, I want to get a complete SEO overview of my website so that I can understand its current performance.
- As a digital marketing analyst, I want to analyze specific SEO elements in detail so that I can identify areas for improvement.

Acceptance Criteria:

- Analysis completes within 30 seconds for most web pages
- All major SEO factors (as listed in section 2.1) are evaluated
- Results are presented in an easy-to-understand format for non-technical users
- Detailed data is available for technical users

Technical Implementation:

- Use Chrome's scripting API to inject content scripts for DOM analysis
- Leverage Chrome's Performance API for Core Web Vitals measurements
- Implement modular architecture for easy addition of new analysis factors

Rationale: A comprehensive analysis provides value to all user personas, from high-level overviews for business owners to detailed insights for SEO professionals.

## 2. Keyword Analysis

- Allow users to input a primary keyword (mandatory field)
- Analyze keyword presence in title, description, and content

Detailed Description: Analyze the presence and usage of a user-specified keyword throughout the web page, including in title, meta description, headers, and content.

User Stories:

- As a small business owner, I want to check if my chosen keyword is properly used on my webpage so that I can improve my chances of ranking for that term.
- As a digital marketing analyst, I want to assess the keyword density and placement on a page so that I can optimize its usage for SEO.

Acceptance Criteria:

- Users can input a primary keyword for analysis.
- The analysis checks for keyword presence in title, meta description, headers (H1-H6), and main content.
- The tool provides a keyword density percentage.
- The analysis highlights instances of keyword stuffing.
- Results include suggestions for improving keyword usage if necessary.

Technical Implementation Details:

- Implement a keyword input field in the extension popup.
- Use regular expressions to identify keyword instances in different page elements.
- Create a function to calculate keyword density (keyword occurrences / total words).
- Implement logic to detect over-optimization (keyword stuffing).
- Use DOM manipulation to highlight keyword instances on the page.

#### Rationale for Design Decisions:

- User input for keywords allows flexibility and relevance for different business needs.
  - Checking multiple page elements ensures comprehensive keyword analysis.
  - Highlighting instances on the page provides visual feedback to users.
  - Detecting keyword stuffing helps users avoid over-optimization penalties.
3. Performance Analysis (Core Web Vitals)

#### 3. Performance Analysis

- Utilize Chrome's PageSpeed API and Lighthouse for performance metrics
- Analyze Core Web Vitals and other important performance indicators

Detailed Description: Measure and analyze the Core Web Vitals (Largest Contentful Paint, First Input Delay, Cumulative Layout Shift) and other key performance metrics of the web page.

#### User Stories:

- As a digital marketing manager, I want to know if my website meets Google's Core Web Vitals requirements so that I can ensure good search engine rankings.
- As a small business owner, I want to understand my website's performance in simple terms so that I can decide if I need to invest in improvements.

#### Acceptance Criteria:

- Accurately measure LCP, FID, and CLS.
- Provide a clear indication of whether each metric meets Google's requirements.
- Include other relevant performance metrics like Time to First Byte (TTFB) and Time to Interactive (TTI).
- Present results in both technical and non-technical language.
- Offer basic suggestions for improving each metric if it doesn't meet the requirements.

#### Technical Implementation Details:

- Use the Performance API and PerformanceObserver to measure Core Web Vitals.
- Implement the chrome.debugger API to gather detailed performance data.
- Create a background script to handle performance measurements across multiple page loads.
- Use Web Vitals library for consistent calculation of metrics.

#### Rationale for Design Decisions:

- Using standard APIs and libraries ensures accuracy and consistency with Google's measurements.
- Background script implementation allows for measurements across page loads, providing more accurate data.
- Dual presentation (technical and non-technical) caters to different user personas.

#### 4. Content Quality Assessment

- Evaluate content structure, readability, and HTML-to-text ratio
- Check for keyword stuffing and thin content

Detailed Description: Evaluate the quality of the page content, including readability, word count, keyword usage, and content structure.

User Stories:

- As a content creator, I want to assess the readability of my content so that I can ensure it's accessible to my target audience.
- As a digital marketing analyst, I want to evaluate the content structure and length so that I can optimize it for SEO.

Acceptance Criteria:

- Calculate and display readability scores (e.g., Flesch-Kincaid).
- Provide word count and compare it to SEO best practices.
- Analyze content structure (headings, paragraphs, lists).
- Check for thin content or duplicate content issues.
- Offer suggestions for content improvement.

Technical Implementation Details:

- Implement readability algorithms (Flesch-Kincaid, SMOG, etc.).
- Use DOM manipulation to extract and analyze content structure.
- Implement functions to detect thin or duplicate content.
- Create a module to generate content improvement suggestions based on the analysis.

Rationale for Design Decisions:

- Multiple readability scores provide a comprehensive view of content accessibility.
- Analyzing content structure helps ensure the content is well-organized for both users and search engines.
- Automated suggestions provide immediate value to users of all expertise levels.

#### 5. Technical SEO Evaluation

- Assess mobile-friendliness using Lighthouse
- Check for proper implementation of canonical URLs, robots.txt, and XML sitemaps

Detailed Description: Assess various technical SEO elements including URL structure, meta tags, site speed, mobile-friendliness, and indexability.

User Stories:

- As a webmaster, I want to check if my website has any technical SEO issues so that I can ensure it's fully optimized for search engines.
- As a digital marketing manager, I want to get an overview of our website's technical SEO health so that I can prioritize improvements.

Acceptance Criteria:

- Check and validate robots.txt and sitemap.xml.
- Analyze URL structure for SEO-friendliness.
- Verify the presence and correctness of canonical tags.
- Assess mobile-friendliness of the page.
- Check for common technical issues (e.g., redirect chains, 404 errors).
- Provide a checklist of passed and failed technical SEO checks.

Technical Implementation Details:

- Use fetch API to check robots.txt and sitemap.xml.
- Implement functions to analyze URL structure and canonical tags.
- Use Chrome's mobile-friendly test API for responsive design check.
- Create a module to check for common technical SEO issues.
- Implement a scoring system for technical SEO health.

Rationale for Design Decisions:

- Comprehensive technical checks cover all major aspects of technical SEO.
- Using Chrome's APIs ensures accurate and up-to-date assessments.
- A checklist format provides clear, actionable insights for users.

Detailed Description: Analyze on-page SEO elements including title tags, meta descriptions, header tags, and image alt texts.

User Stories:

- As a content manager, I want to ensure all our pages have optimized titles and descriptions so that they perform well in search results.
- As an SEO specialist, I want to check if our pages are using header tags correctly so that I can improve our content structure.

Acceptance Criteria:

- Verify presence and optimize length of title tags and meta descriptions.
- Check for proper use of header tags (H1, H2, etc.).
- Analyze image alt texts for presence and relevance.
- Identify any missing or duplicate on-page elements.
- Provide suggestions for improving on-page elements.

#### Technical Implementation Details:

- Use DOM manipulation to extract and analyze on-page elements.
- Implement functions to check element presence, length, and structure.
- Create algorithms to assess the quality and relevance of elements.
- Develop a module to generate improvement suggestions.

#### Rationale for Design Decisions:

- Focusing on key on-page elements ensures coverage of the most impactful SEO factors.
- Automated analysis and suggestions save time for users of all expertise levels.
- Quality assessment of elements (not just presence) provides deeper insights.

### 6. On-Page Element Analysis

- Evaluate title tags, meta descriptions, and URL structure
- Check for proper header tag (H1-H6) usage

Detailed Description: Analyze on-page SEO elements including title tags, meta descriptions, header tags, and image alt texts.

#### User Stories:

- As a content manager, I want to ensure all our pages have optimized titles and descriptions so that they perform well in search results.
- As an SEO specialist, I want to check if our pages are using header tags correctly so that I can improve our content structure.

#### Acceptance Criteria:

- Verify presence and optimize length of title tags and meta descriptions.
- Check for proper use of header tags (H1, H2, etc.).
- Analyze image alt texts for presence and relevance.
- Identify any missing or duplicate on-page elements.
- Provide suggestions for improving on-page elements.

#### Technical Implementation Details:

- Use DOM manipulation to extract and analyze on-page elements.
- Implement functions to check element presence, length, and structure.

- Create algorithms to assess the quality and relevance of elements.
- Develop a module to generate improvement suggestions.

Rationale for Design Decisions:

- Focusing on key on-page elements ensures coverage of the most impactful SEO factors.
- Automated analysis and suggestions save time for users of all expertise levels.
- Quality assessment of elements (not just presence) provides deeper insights.

## 7. Media Optimization Check

- Analyze image alt text, size, and format
- Check for proper implementation of lazy loading

Detailed Description: Analyze and evaluate the optimization of images, videos, and other media elements on the webpage for SEO purposes.

User Stories:

- As a content creator, I want to ensure all images on my website are optimized so that they don't slow down page loading times.
- As a digital marketing manager, I want to check if our media elements are SEO-friendly so that they contribute positively to our search engine rankings.

Acceptance Criteria:

- Check image file sizes and formats.
- Verify the presence and quality of alt text for images.
- Analyze video embedding and ensure proper markup.
- Check for responsive images.
- Provide recommendations for media optimization.

Technical Implementation Details:

- Use DOM manipulation to identify all media elements on the page.
- Implement functions to check image file sizes and formats.
- Create algorithms to assess the quality of alt text.
- Use Chrome's network information API to evaluate media loading times.
- Develop a module to generate media optimization suggestions.

Rationale for Design Decisions:

- Focusing on media optimization addresses both SEO and user experience concerns.
- Automated checks save time for content creators and marketers.



- Providing specific recommendations helps users take immediate action to improve their media elements.

## 8. User Experience and Accessibility Evaluation

- Assess page accessibility using Chrome's Accessibility API
- Evaluate CSS complexity and JavaScript usage

Detailed Description: Assess the website's user experience and accessibility features, which indirectly impact SEO through user engagement metrics.

User Stories:

- As a website owner, I want to ensure my site is accessible to all users so that I can maximize my audience reach and comply with accessibility standards.
- As a UX designer, I want to identify any user experience issues that might negatively impact our SEO performance.

Acceptance Criteria:

- Evaluate website navigation and structure.
- Check for proper use of ARIA labels and roles.
- Assess color contrast ratios for text readability.
- Verify keyboard navigation functionality.
- Provide a checklist of passed and failed accessibility checks.

Technical Implementation Details:

- Use `chrome.accessibility` API to gather accessibility information.
- Implement functions to analyze site structure and navigation patterns.
- Create algorithms to check color contrast ratios.
- Develop tests for keyboard navigation.
- Use Web Content Accessibility Guidelines (WCAG) as a benchmark for checks.

Rationale for Design Decisions:

- Incorporating accessibility checks acknowledges the growing importance of inclusive web design.
- UX evaluation recognizes the indirect impact of user experience on SEO performance.
- Using established guidelines (WCAG) ensures compliance with industry standards.

## 9. Link Analysis

- Analyze internal and external links

- Check for broken links

Detailed Description: Analyze internal and external links on the webpage, including their quantity, quality, and relevance.

User Stories:

- As an SEO specialist, I want to review our website's link structure so that I can optimize our internal linking strategy.
- As a content manager, I want to ensure our external links are high-quality and relevant so that they add value to our content and SEO.

Acceptance Criteria:

- Count and categorize internal and external links.
- Check for broken links.
- Analyze anchor text relevance.
- Evaluate the distribution of link juice across the site.
- Provide recommendations for improving link structure.

Technical Implementation Details:

- Use DOM manipulation to extract all links from the page.
- Implement functions to categorize links as internal or external.
- Create algorithms to assess anchor text relevance.
- Use fetch API to check for broken links.
- Develop a module to analyze link juice distribution based on page structure.

Rationale for Design Decisions:

- Comprehensive link analysis helps improve both user experience and SEO.
- Automated broken link checking saves time and improves site reliability.
- Analyzing link juice distribution helps optimize the site's overall SEO strategy.

## 10. Structured Data Validation

- Check for the presence and validity of structured data markup

Detailed Description: Check for the presence and validity of structured data (schema markup) on the webpage.

User Stories:

- As a webmaster, I want to verify that our structured data is correctly implemented so that search engines can better understand our content.

- As an SEO manager, I want to identify opportunities to add structured data to our pages so that we can enhance our search result appearances.

#### Acceptance Criteria:

- Detect presence of structured data (JSON-LD, Microdata, RDFa).
- Validate structured data against schema.org specifications.
- Identify missing or incomplete structured data.
- Suggest additional structured data types that could be beneficial.
- Provide a summary of structured data findings and recommendations.

#### Technical Implementation Details:

- Use DOM manipulation to extract structured data from the page.
- Implement JSON-LD, Microdata, and RDFa parsers.
- Create validation functions based on schema.org specifications.
- Develop a recommendation system for suggesting additional structured data types.
- Use visualization libraries to display structured data in a user-friendly format.

#### Rationale for Design Decisions:

- Supporting multiple structured data formats ensures comprehensive analysis.
- Validation against schema.org specs helps ensure compliance with search engine requirements.
- Suggesting additional structured data types provides added value to users.

### 11. Security Assessment

- Verify HTTPS implementation

Detailed Description: Evaluate basic security features of the webpage that can impact SEO, such as HTTPS implementation and safe browsing status.

#### User Stories:

- As a website owner, I want to ensure my site is secure so that it doesn't negatively impact my search engine rankings.
- As an SEO analyst, I want to check for any security issues that might be affecting our SEO performance.

#### Acceptance Criteria:

- Verify HTTPS implementation and SSL certificate validity.
- Check for mixed content issues.
- Assess implementation of security headers.
- Verify the site's status in Google Safe Browsing.
- Provide a security checklist with passed and failed checks.

#### Technical Implementation Details:

- Use chrome.webRequest API to analyze HTTPS implementation and security headers.
- Implement functions to detect mixed content.
- Create API integration with Google Safe Browsing (if possible within extension limitations).
- Develop a module to generate security recommendations based on failed checks.

#### Rationale for Design Decisions:

- Focus on security aspects that directly impact SEO.
- Automated security checks provide value to users who may not have security expertise.
- Integration with Google Safe Browsing (if possible) provides authoritative security status information.

#### 12. Compliance Scoring

- Evaluate GDPR, CCPA, and ADA compliance

##### Detailed Description:

Evaluate the website's compliance with key regulations such as GDPR, CCPA, and ADA, which can indirectly affect SEO through user trust and accessibility.

##### User Stories:

- As a website owner, I want to ensure my site complies with major regulations so that I can avoid potential legal issues and improve user trust.
- As a digital marketing manager, I want to assess our website's compliance status so that I can report on potential risks and improvement areas to stakeholders.

##### Acceptance Criteria:

- Provide a compliance score for GDPR, CCPA, and ADA separately.
- Check for presence of privacy policy and cookie consent mechanisms.
- Evaluate accessibility features for ADA compliance.
- Offer a checklist of compliance factors for each regulation.
- Provide recommendations for improving compliance.

##### Technical Implementation Details:

- Develop algorithms to check for common compliance indicators (e.g., privacy policy links, cookie notifications).
- Use the chrome.accessibility API to assess ADA compliance factors.
- Create a scoring system that weighs different compliance factors.
- Implement a module to generate compliance improvement suggestions.
- Use legal databases or APIs to keep compliance checks up-to-date with current regulations.

Rationale for Design Decisions:

- Addressing compliance indirectly supports SEO by improving user trust and accessibility.
- Separate scores for each regulation allow for more targeted improvements.
- Automated compliance checks provide value to users who may not have legal expertise in these areas.

### 13. Actionable Recommendations

- Provide specific, prioritized recommendations based on the analysis

Detailed Description: Provide specific, prioritized recommendations for improving the website's SEO based on the analysis results.

User Stories:

- As a small business owner, I want to receive clear, actionable advice on how to improve my website's SEO so that I can make meaningful changes without extensive technical knowledge.
- As an SEO specialist, I want to get detailed recommendations for SEO improvements so that I can create a comprehensive optimization plan.

Acceptance Criteria:

- Generate specific recommendations based on the analysis results.
- Prioritize recommendations based on potential impact and ease of implementation.
- Provide both quick wins and long-term strategies.
- Include brief explanations of why each recommendation is important.
- Offer guidance on how to implement each recommendation.

Technical Implementation Details:

- Create a knowledge base of common SEO issues and corresponding recommendations.
- Develop an algorithm to match analysis results with appropriate recommendations.
- Implement a prioritization system based on impact and difficulty scores.
- Design a user-friendly interface to display recommendations in an easy-to-understand format.

Rationale for Design Decisions:

- Actionable recommendations provide immediate value to users of all expertise levels.
- Prioritization helps users focus on the most impactful changes first.
- Including both quick wins and long-term strategies caters to different user needs and resources.
- Explanations and implementation guidance educate users and increase the likelihood of recommendations being acted upon.

#### 14. CSV Export Feature

- Allow users to download a detailed SEO report in CSV format

Detailed Description: Allow users to export a detailed SEO analysis report in CSV format for further analysis or reporting purposes.

User Stories:

- As an SEO specialist, I want to export the full SEO analysis data so that I can create custom reports or perform additional analysis in spreadsheet software.
- As a digital marketing manager, I want to download SEO reports in CSV format so that I can easily share the data with my team or clients.

Acceptance Criteria:

- Provide an option to export the full SEO analysis report as a CSV file.
- Include all relevant metrics, scores, and recommendations in the export.
- Ensure the CSV is well-formatted and easy to read in spreadsheet software.
- Allow users to choose which sections of the report to include in the export.
- Implement the export feature without significantly impacting the extension's performance.

Technical Implementation Details:

- Develop a module to format analysis data into CSV structure.
- Use the chrome.downloads API to handle file downloads.
- Implement data filtering options to allow customization of exported content.
- Ensure proper encoding to handle special characters and different languages.
- Optimize the export process to handle large amounts of data efficiently.

Rationale for Design Decisions:

- CSV format is widely supported and can be easily imported into various analysis tools.
- Allowing customization of exported data increases the feature's flexibility and usefulness.
- Using chrome.downloads API ensures smooth integration with the browser's download functionality.

#### 15. Industry Benchmarks for Comparison

Detailed Description: Provide industry-specific benchmarks to compare the website's SEO performance against similar sites in the same sector.

User Stories:

- As a digital marketing manager, I want to see how our website's SEO performance compares to industry standards so that I can set realistic improvement goals.
- As a small business owner, I want to understand if my SEO efforts are on par with my competitors so that I can determine if I need to invest more in SEO.

#### Acceptance Criteria:

- Offer a selection of industry categories for users to choose from.
- Provide benchmark data for key SEO metrics within each industry.
- Compare the analyzed website's performance against these benchmarks.
- Highlight areas where the site is above or below industry standards.
- Offer insights into what top performers in the industry are doing differently.

#### Technical Implementation Details:

- Create a database of industry benchmarks based on aggregated anonymous data or third-party sources.
- Implement a system for users to select their industry category.
- Develop comparison algorithms to match site metrics against benchmarks.
- Use data visualization libraries to present benchmark comparisons graphically.

#### Rationale for Design Decisions:

- Industry-specific benchmarks provide more relevant and actionable insights than generic standards.
- Allowing users to select their industry ensures accurate comparisons.
- Visual representation of benchmarks makes it easy for users to quickly understand their position in the market.

Certainly. Here are the detailed specifications for the P2 (Nice-to-have) features:

### 16. Social Media Assessment

#### Detailed Description:

Analyze the integration and optimization of social media elements on the website, including share buttons, Open Graph tags, and Twitter Card markup.

#### User Stories:

- As a social media manager, I want to ensure our website is optimized for social sharing so that our content performs well when shared on social platforms.
- As a digital marketer, I want to check if our social media integration is supporting our SEO efforts effectively.

#### Acceptance Criteria:

- Check for the presence and proper implementation of social share buttons.
- Verify the presence and correctness of Open Graph tags.
- Analyze Twitter Card markup.

- Evaluate the loading speed of social media widgets.
- Provide recommendations for improving social media integration.

#### Technical Implementation Details:

- Use DOM manipulation to identify social media elements on the page.
- Implement functions to validate Open Graph and Twitter Card markup.
- Create algorithms to assess the impact of social media widgets on page load speed.
- Develop a module to generate social media optimization suggestions.
- Consider integration with social media APIs for more in-depth analysis (within browser extension limitations).

#### Rationale for Design Decisions:

- Social media optimization indirectly supports SEO through increased engagement and backlinks.
- Automated checks for social markup save time for marketers and developers.
- Assessing the performance impact of social widgets helps balance functionality with page speed.

### 17. Historical Tracking of SEO Scores

#### Detailed Description:

Allow users to track changes in their SEO scores over time, providing insights into the effectiveness of their optimization efforts.

#### User Stories:

- As an SEO specialist, I want to see how our SEO scores have changed over time so that I can measure the impact of our optimization efforts.
- As a digital marketing manager, I want to track our SEO progress to report improvements to stakeholders.

#### Acceptance Criteria:

- Store historical SEO scores for each analyzed URL.
- Display a graph showing score changes over time.
- Allow users to select date ranges for viewing historical data.
- Highlight significant changes in scores and potential reasons for these changes.
- Provide an option to export historical data.

#### Technical Implementation Details:

- Use `chrome.storage.local` to store historical score data.
- Implement data structures to efficiently store and retrieve time-series data.
- Use a charting library (e.g., `Chart.js`) to visualize historical trends.
- Develop algorithms to identify significant changes in scores.
- Create a module to manage data retention and prevent excessive storage usage.

#### Rationale for Design Decisions:

- Historical tracking provides valuable insights into the effectiveness of SEO efforts over time.



- Visualization of trends makes it easy for users to understand progress at a glance.
- Local storage ensures quick access to historical data without relying on external servers.

## 18. Competitive Analysis Feature

### Detailed Description:

Allow users to compare their website's SEO performance against competitors, providing insights into relative strengths and weaknesses.

### User Stories:

- As an SEO manager, I want to see how our website's SEO compares to our main competitors so that I can identify areas where we need to improve.
- As a business owner, I want to understand our SEO performance relative to competitors so that I can make informed decisions about resource allocation.

### Acceptance Criteria:

- Allow users to input URLs of competitor websites for comparison.
- Provide side-by-side comparisons of key SEO metrics.
- Highlight areas where the user's website is stronger or weaker than competitors.
- Offer insights into potential strategies based on competitive analysis.
- Ensure the feature respects robots.txt and doesn't violate any terms of service.

### Technical Implementation Details:

- Develop a UI for inputting and managing competitor URLs.
- Create algorithms to perform SEO analysis on competitor sites (within ethical and legal bounds).
- Implement data structures to store and compare metrics across multiple sites.
- Use visualization libraries to present comparative data in an easy-to-understand format.
- Ensure proper error handling for cases where competitor sites can't be analyzed.

### Rationale for Design Decisions:

- Competitive analysis provides context for SEO performance and helps in strategy formation.
- Side-by-side comparisons make it easy to identify relative strengths and weaknesses.
- Ethical considerations are paramount to ensure the feature doesn't engage in any questionable data collection practices.

## 19. Custom Scoring Options for Users

### Detailed Description:

Allow users to customize the weighting of different SEO factors in the overall score calculation, tailoring the analysis to their specific priorities or industry needs.

### User Stories:

- As an SEO specialist, I want to adjust the importance of different SEO factors in the scoring system so that it aligns with my specific strategy or industry requirements.

- As a digital marketing manager, I want to customize our SEO scoring to focus on the metrics that are most important for our business goals.

#### Acceptance Criteria:

- Provide an interface for users to adjust the weight of different SEO factors.
- Allow saving of custom scoring profiles.
- Recalculate scores based on custom weightings.
- Offer pre-set scoring profiles for common industry types.
- Provide an option to reset to default scoring.

#### Technical Implementation Details:

- Develop a user interface for adjusting factor weights.
- Implement a flexible scoring algorithm that can incorporate custom weights.
- Use `chrome.storage.sync` to save and retrieve custom scoring profiles.
- Create a system for managing and applying pre-set industry profiles.
- Develop safeguards to prevent users from creating invalid scoring configurations.

#### Rationale for Design Decisions:

- Customizable scoring acknowledges that SEO priorities can vary by industry and business goals.
- Saved profiles allow users to quickly switch between different scoring models.
- Pre-set industry profiles provide value for users who may not know how to optimally set weights themselves.
- Synced storage allows users to access their custom profiles across devices.

## 20. Scoring System

The scoring system is based on a 100-point scale, divided into the following categories:

1. Content Quality (25% of total score)
2. Technical SEO (25% of total score)
3. On-Page Elements (20% of total score)
4. Media and Formatting (10% of total score)
5. User Experience and Accessibility (10% of total score)
6. Links and Structured Data (5% of total score)
7. Security and Social (5% of total score)
8. Compliance (5% of total score)

Each factor within these categories is assigned a severity level:

- Critical
- High
- Medium
- Low

Performance Tiers:

- Excellent: 90-100
- Good: 75-89
- Needs Improvement: 60-74
- Poor: 0-59

For each factor, the extension will provide:

- Score Explanation: A brief description of why the score was given and its importance to SEO.
- Action Items: Specific, actionable recommendations for improvement.

Industry benchmarks will be provided for each major category based on aggregated data from various SEO studies and best practices.

## 3. Non-Functional Requirements

### 3.1 Usability

- User Interface: Simple and intuitive UI within the Chrome extension popup.
- Accessibility: Ensure the extension is accessible and usable for individuals with disabilities.

### 3.2 Performance

- Fast Analysis: Perform SEO analysis quickly without significant lag.
- Efficient Resource Usage: Minimize memory and CPU usage during operation.
  - Maximum analysis time: 30 seconds
  - Maximum memory usage: 100MB
  - Maximum CPU usage: 50% for no more than 5 seconds

### 3.3 Security

- Data Privacy: Ensure user data and analyzed content are kept private and secure.
- Secure Storage: Use secure storage for saving user settings and historical analyses.

## 4. Technical Specifications

### 4.1 Technologies

- JavaScript: For core logic and interaction with the web pages.
- HTML/CSS: For the extension's user interface.
- Chrome Extension API: To interact with Chrome and manipulate web pages.
- Chrome PageSpeed API: For performance analysis.
- Chrome Accessibility API: For accessibility evaluation.
- Storage API: To persist user settings and previous analysis data.

## 4.2 Dependencies

- Chrome APIs: To access active tab, storage, and background tasks.
- Lighthouse: For performance and mobile-friendliness analysis.
- Web Fetch API: To validate links' status.

Chrome APIs to be used:

```
chrome.tabs
chrome.storage
chrome.runtime
chrome.scripting
chrome.debugger (for performance analysis)
```

Example API call:

```
chrome.tabs.query({active: true, currentWindow: true}, function(tabs) {
  let currentTab = tabs[0];
  chrome.tabs.sendMessage(currentTab.id, {action: "analyze"}, function(response) {
    // Handle response
  });
});
```

## 4.3 Data Structures

Page Analysis Result Object:

```
````javascript
{
  url: string,
  timestamp: number,
  overallScore: number,
  categories: {
    contentQuality: {
      score: number,
      factors: [
        {
          name: string,
          score: number,
          severity: 'critical' | 'high' | 'medium' | 'low',
          explanation: string,
          recommendation: string
        }
      ]
    },
    // Similar structure for other categories
  },
  performanceTier: 'Excellent' | 'Good' | 'Needs Improvement' | 'Poor'
}
```

## 4.4 Algorithms

### Scoring System Algorithm:

```
function calculateOverallScore(categories):
    totalScore = 0
    for each category in categories:
        categoryScore = 0
        for each factor in category.factors:
            factorScore = factor.score * factor.weight
            categoryScore += factorScore
        totalScore += categoryScore * category.weight
    return totalScore

function determinePerformanceTier(score):
    if score >= 90: return 'Excellent'
    else if score >= 75: return 'Good'
    else if score >= 60: return 'Needs Improvement'
    else: return 'Poor'
```

## 5. Detailed Components

### 5.1 Manifest File (manifest.json)

- Define the extension's permissions, background scripts, content scripts, and popup UI.

### 5.2 Popup HTML (popup.html)

- Structure for interaction with the user, including input fields and buttons.

### 5.3 Popup CSS (popup.css)

- Styling for the extension's user interface.

### 5.4 Content Script (content.js)

- Logic to analyze the page, gather data, and return results.

### 5.5 Popup JS (popup.js)

- Event handling for user actions, such as initiating page analysis and exporting reports.

### 5.6 Background Script (background.js)

- Handle background tasks, such as initializing storage and exporting reports.
- Implement the scoring system and benchmark comparisons.

## 6. Error Handling and Logging

- Implement comprehensive error handling in all scripts to capture and log errors.

- Display errors in the chrome extension popup when encountered.
- Ensure that errors do not crash the extension but provide meaningful feedback to the user.
- Implement proper error handling for API calls and data processing.

## 7. User Workflow

1. Initialization: On first install, set up default settings and storage structures.
2. Page Analysis:
  - User opens the popup.
  - User inputs the primary keyword (mandatory).
  - User clicks "Analyze Page".
  - Content script performs the analysis and sends results back to the popup.
  - Popup displays the overall SEO score, category scores, and performance tier.
3. Detailed Results:
  - User can view detailed breakdowns for each category.
  - Each factor shows its score, explanation, and actionable recommendations.
  - Industry benchmarks are displayed for comparison.
4. Export Report:
  - User clicks "Download Detailed Report (CSV)".
  - The report is generated and downloaded, including all metrics, issues, and recommendations.

## 8. Testing and Quality Assurance

- Unit Testing: Write unit tests for individual functions, including scoring calculations.
- Integration Testing: Ensure all components work seamlessly together.
- User Testing: Gather feedback from users and make necessary adjustments.
- Performance Testing: Ensure the extension operates efficiently on various page types and sizes.
- Cross-version Testing: Test on different versions of Chrome to ensure compatibility.

## 9. Maintenance and Updates

- Version Updates: Provide regular updates to improve performance and add new features.
- Bug Fixes: Address user-reported bugs promptly.
- Algorithm Updates: Regularly update the scoring system and benchmarks to align with the latest SEO best practices.

## 10. Browser Compatibility

- Follow Google's developer recommendations for Chrome extension compatibility.
- Clearly specify the minimum supported Chrome version in the extension description.

## 11. Security Considerations

- Follow Google's developer recommendations for Chrome extension security.
- Implement secure coding practices to prevent vulnerabilities.
- Ensure user data privacy and secure handling of analyzed content.

Conclusion: This SEO Analyzer Chrome Extension aims to be a comprehensive and user-friendly tool for webmasters, marketers, and SEO enthusiasts to optimize their web pages. With its detailed analysis, actionable recommendations, and user-friendly interface, users can significantly enhance their website's search engine performance.

## 12. Change Log

### **Version: 3.2 (Current)**

Release Date: 29th July 2024

### **New Features:**

1. Progress Bar: Added a real-time progress bar to provide visual feedback during the analysis process.
2. Collapsible Category Scores: Implemented a toggle feature to collapse/expand category scores for better user experience.
3. Enhanced Performance Analysis: Expanded Core Web Vitals metrics to include FCP, TTI, and TBT.

### **Improvements:**

1. Updated UI: Redesigned the results summary section for better clarity and information hierarchy.

2. Enhanced CSV Export: Added more detailed performance metrics and Core Web Vitals to the exported report.
3. Improved Error Handling: Implemented more user-friendly error messages in the extension popup.

#### **Bug Fixes:**

1. Fixed an issue where the analysis sometimes failed on pages with complex DOM structures.
2. Resolved a memory leak that occurred during repeated analyses.
3. Corrected the calculation of the overall SEO score to more accurately reflect individual category scores.

#### **API Changes:**

1. Updated to use the latest version of Chrome's PageSpeed Insights API for more accurate performance data.
2. Implemented the chrome.downloads API for improved CSV export functionality.

#### **Documentation Updates:**

1. Updated the user guide to reflect new features and UI changes.
2. Added more detailed API documentation for developers looking to extend the extension's functionality.

#### **Known Issues:**

1. Analysis may take longer than expected on very large web pages (investigating performance optimizations for future release).
2. The extension may not function correctly on some non-standard webpage implementations (ongoing compatibility improvements).

#### **Deprecations:**

1. Removed support for older Chrome versions (minimum supported version is now [Insert Version Number]).



#### Future Plans:

1. Investigating the implementation of a competitive analysis feature.
2. Considering the addition of custom scoring options for users with specific industry needs.

This changelog provides a comprehensive overview of the changes, improvements, and future plans for the SEO Analyzer Chrome Extension. It helps users and stakeholders understand what's new, what's been fixed, and what to expect in future updates.